Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
Protocols Involved: UDP (used by DNS queries), ICMP (used for error reporting) **Analysis:** The tcpdump log shows that the client (Source IP: 192.51.100.15, Source Port: 52444) send a DNS query to the server (Destination IP: 203.0.113.2) on UDP port 53. **Log Details:** The server responded with an ICMP "UDP Port 53 Unreachable length 254". **Interpretation:** This indicates that the DNS service on the server is not listening on port 53, meaning DNS queries cannot be processed. **Most Likely Issues:** - DNS server service is down - UDP port 53 is closed or not listening **Impact on Users:** As a result, clients cannot resolve the domain yummyrecipesforme.com and cannot access the website.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time of Incident: 13:24 – 13:28 **How IT team became aware:** Customers reported they could not access www.yummyrecipresforme.com and received "Destination Port Unreachable" errors. **ICMP Explanation:** "Destination Unreachable" message is a feedback to the client that the requested service on UDP port 53 is not available. **Investigation Actions:** - Loaded tcpdump on the network to capture traffic - Identified ICMP messages indicating UDP port 53 was unreachable - Confirmed that the DNS queries reached the server, but the server did not respond because the DNS service was down or port 53 was closed.

Current Status: The website is down; DNS queries cannot be resolved.

Impact: All clients attempting to access the website experienced DNS failure, resulting in downtime.

Next Steps:
1. Check the status of the DNS Service and ensure the DNS server is running (*systemctl status named or bind9*).
2. Check if UDP port 53 is open/listening and confirm if the port is listening (*netstat -tulnp | grep 53 or ss -u -l -n*).
3. Start or restart the DNS Service.
4. Test domain resolution (*dig yummyrecipesforme.com or nslookup yummyrecipesforme.com*).
5. Check the website and attempt to access the website to confirm issue is resolved.
6. Document findings and resolution steps.

Suspected root cause:
- DNS Service Down
– UDP port 53 was not listening (closed) at the time of the incident